

Spatial Machine Learning in R

Obtaining information from the
environment

Session 3

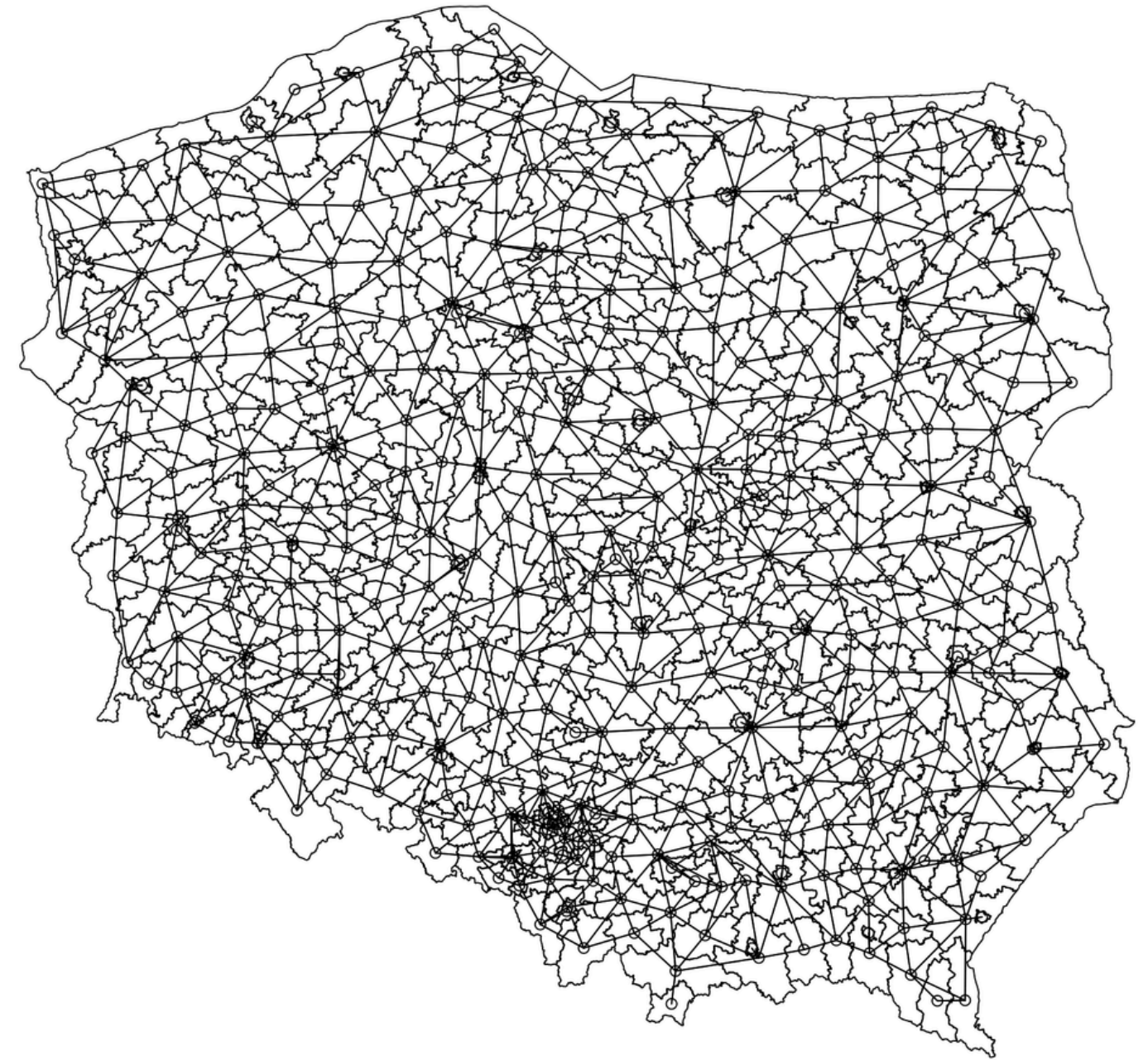
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Obtaining information from the environment

- Observations do not exist in isolation
- What happens around a unit often matters
- Today: how to formally define and measure “surroundings”

Key questions:

- Who influences whom?
- How far does influence reach?
- How do we measure neighbourhood effects?



From location to interaction

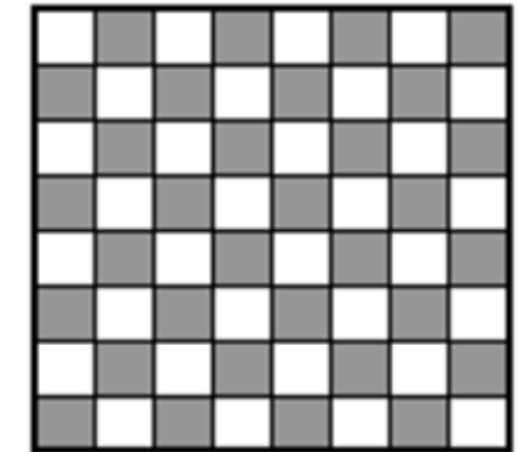
Problem

- Standard models assume that observations are independent
- But in space, nearby regions share shocks, firms influence each other
- Spatial data violate independence by default

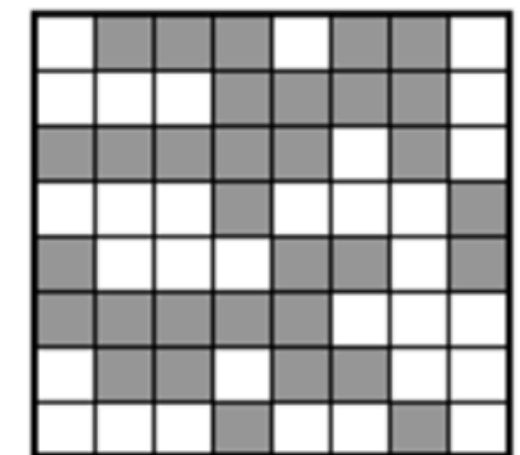
Solution

We explicitly model spatial relationships

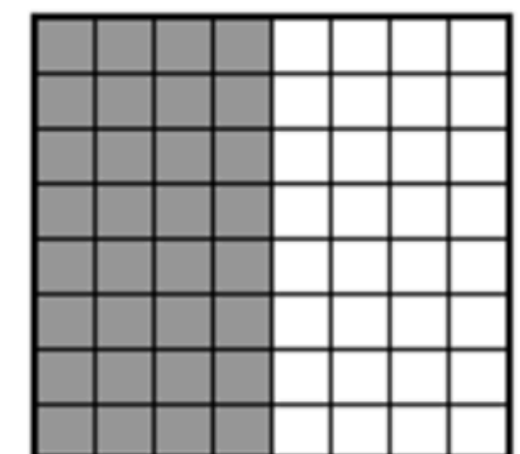
Negative autocorrelation



Random distribution



Positive autocorrelation



What is a Spatial Weights Matrix (W)?

A spatial weights matrix W :

$$W = [w_{ij}]_{n \times n}$$

$$w_{ij} = \begin{cases} > 0 & \text{if units } i \text{ and } j \text{ are neighbours} \\ 0 & \text{otherwise} \end{cases}$$

$$w_{ii} = 0 \quad \text{for all } i$$

$$W = \begin{pmatrix} w_{11} & w_{12} & \cdots & w_{1n} \\ w_{21} & w_{22} & \cdots & \vdots \\ \vdots & \vdots & w_{ii} & \vdots \\ w_{n1} & \cdots & \cdots & w_{nn} \end{pmatrix}$$

Two crucial steps:

1. Define neighbours (structure)
2. Define weights (strength + standardisation)

$$W = \begin{pmatrix} 0 & 1 & 0 & 1 & 0 \\ 1 & 0 & 1 & 1 & 1 \\ 0 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 & 0 \end{pmatrix}$$

$$W^* = \begin{pmatrix} 0 & 0.5 & 0 & 0.5 & 0 \\ 0.25 & 0 & 0.25 & 0.25 & 0.25 \\ 0 & 0.5 & 0 & 0 & 0.5 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0.5 & 0 & 0 \end{pmatrix}$$

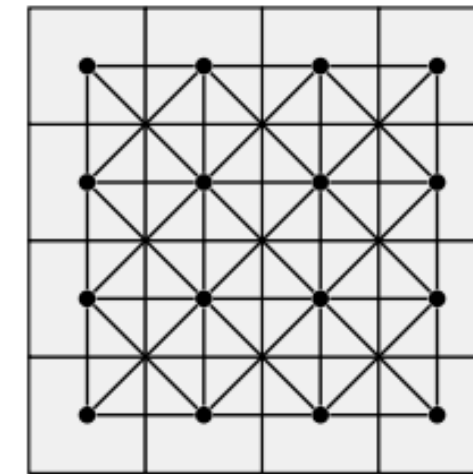
$$\sum_{j=1}^n w_{ij}^* = 1$$

Different ways to define neighbourhood

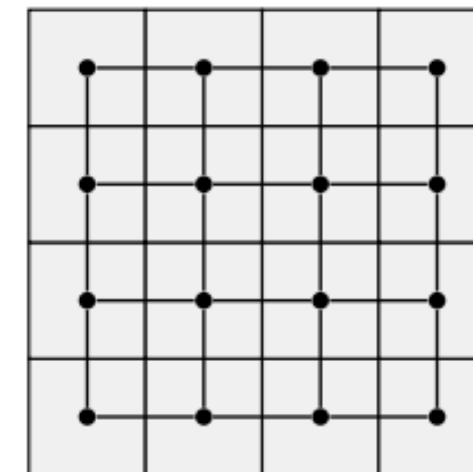
How do we define neighbours in spatial weight matrices?

Some of the options:

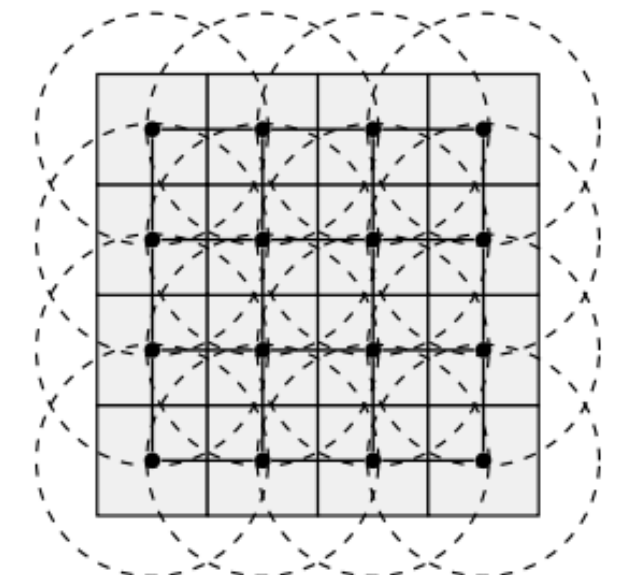
- **Contiguity (Polygons):** share border → neighbour
- **KNN:** k closest units are neighbours
- **Distance band:** if unit is within radius d then it's a neighbour
- **Inverse distance:** everyone influences everyone, but with decay
- **Voronoi (Points → Polygons):** artificial boundaries for point data



Queen contiguity



k -NN ($k = 2$)



Distance band

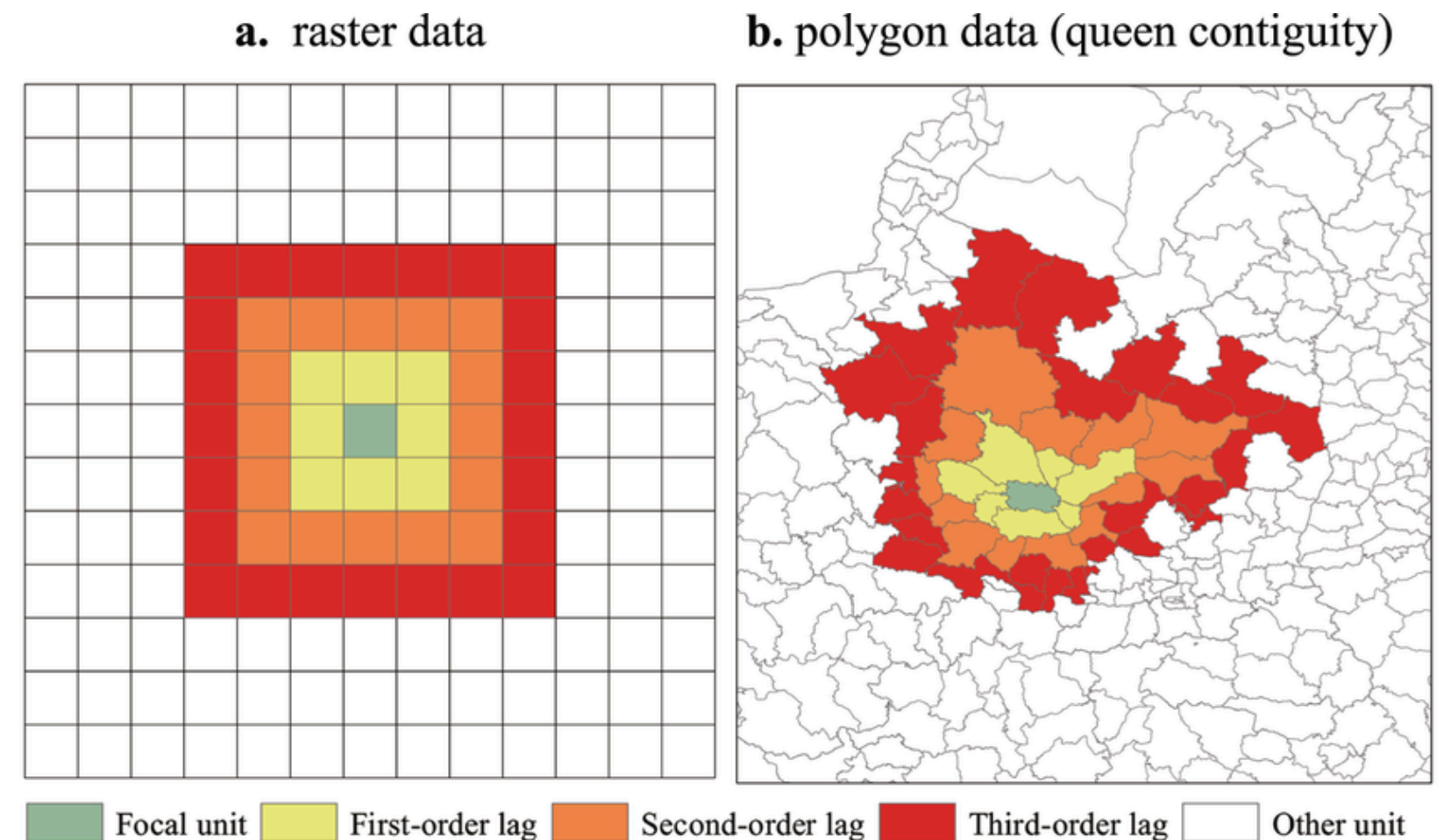
Spatial lags: bringing environment into the variable

Spatial lag = environmental summary

- A spatial lag answers the question:
“What is happening around me?”
- It summarises the values of neighbouring units
- It uses the spatial weights matrix W
- It becomes a weighted local average (if row-standardised)

$$(Wy)_i = \sum_{j=1}^n w_{ij}y_j \quad \text{for each unit } i$$

Spatial lag of y_i = weighted average of neighbours' y_j

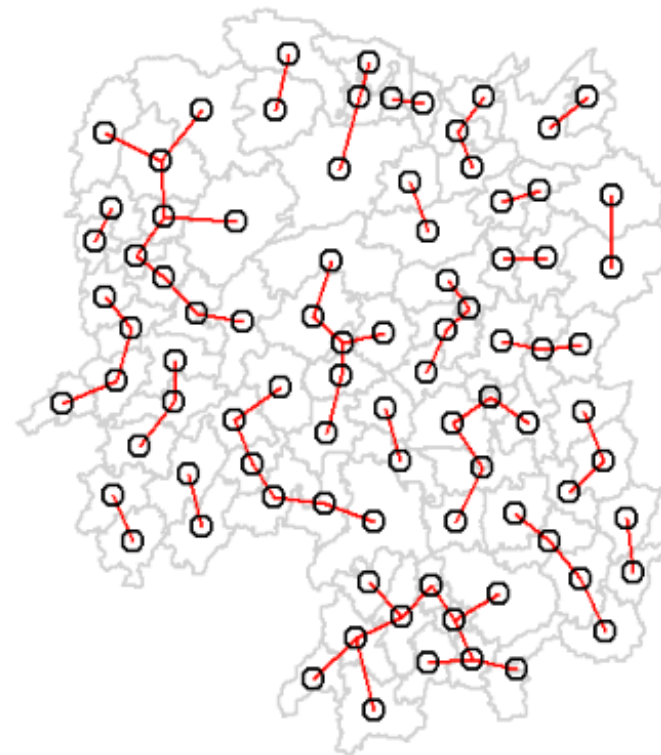


Source: https://www.researchgate.net/figure/Spatial-lags-a-The-spatial-focal-unit-and-its-spatial-lags-in-different-orders-of-raster_fig1_374085389

When can spatial weights be problematic?

1. **Results depend on W** - Contiguity or distance matrix? What should be the k or distance band? Different spatial weights matrices can produce different model results.
2. **Fragmentation** - Some units might have no neighbours. Fragmented networks weaken diffusion and spatial feedback.
3. **Heavy computations** with many units/points

Links of 1st nearest neighbour



Neighbours within 62 km



Source: <https://rpubs.com/erikaaldisa/spatialweights>

From regional to micro perspective

Regional W matrices:

- Work well for administrative units

But for points (e.g. firms):

- No borders
- Interactions are highly local
- Environment varies continuously

Solution

- Radial zoning
- Buffers
- Local counts

